

Report
Of
Game Development
Fruit Merge – Melon Maker and Ball Sort

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By
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Roll No.: CS-2341524
Batch: 2023-27

CANDIDATE'S DECLARATION

I, Ankit Kumar, do hereby declare that the internship report titled “Game Development: Fruit Merge – Melon Maker and Ball Sort” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references used in preparing this report have been acknowledged, and the report presents the work carried out during my internship with IBRA Digital Branding Services Pvt. Ltd.

I affirm that this report is my own work and has not been submitted to any other institute for any degree or diploma requirement. I shall be solely responsible for any copyright violation, if any, arising from material submitted by me.

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their guidance, constructive criticism and advice during the work. I am sincerely grateful to the people who shared their knowledge and practical views related to game development and the completion of this internship.

I express my gratitude to IBRA Digital Branding Services Pvt. Ltd. for providing me the opportunity to work as a Game Developer Intern in the Game Development Department. The internship provided practical exposure to mobile game development, testing, debugging and preparation of Android/Play-Store-ready builds.

I also express my gratitude to my parents for their blessings and continuous support.

Student Name: ANKIT KUMAR

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INTERNSHIP COMPLETION CERTIFICATE



Ibra Digital Branding Services Pvt. Ltd.

www.ibradigitals.com

CIN: U72900DL2020PTC364934

PAN: ABDCS7918N

GSTIN (Delhi): 07ABDCS7918N12P

Date: 15-Aug-2026

TO WHOM IT MAY CONCERN

This is to certify that **Ankit Kumar** successfully completed his internship with IBRA Digital Branding Services Pvt. Ltd. from **15 April 2026** to **15 August 2026** as a **Game Developer Intern** in the Game Development Department.

During his internship, he worked on the development of mobile games for **Android platforms**, including game functionality, testing, debugging, and preparing Play-Store ready builds for publication.

He demonstrated good technical skills, creativity, and commitment throughout his internship.

We wish him all the very best for his career.

Asad Rabbani
CEO

For
IBRA Digital Branding Services Pvt. Ltd.


Director

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Certificate issued by IBRA Digital Branding Services Pvt. Ltd.

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4. PROJECT DESCRIPTION

4.1 Introduction

This report presents the work carried out during my internship as a Game Developer Intern at IBRA Digital Branding Services Pvt. Ltd. in the Game Development Department. The internship was completed from 15 April 2026 to 15 August 2026. As stated in the internship completion certificate, the work involved development of mobile games for Android platforms, including game functionality, testing, debugging and preparation of Play-Store-ready builds.

The two game projects documented in this report are Fruit Merge – Melon Maker and Ball Sort. Both projects are mobile-oriented puzzle games developed using GDevelop. The work involved implementing gameplay functionality, organizing layouts and game objects, using event-driven logic, integrating visual and audio assets, testing interactions and refining the overall player experience.

4.2 Organization Profile

IBRA Digital Branding Services Pvt. Ltd. is the organization where the internship was completed. The internship was carried out in the Game Development Department under the designation Game Developer Intern. The internship certificate confirms the duration from 15 April 2026 to 15 August 2026 and describes the work as mobile-game development for Android platforms.

The internship provided an industry-oriented environment in which game-development tasks were approached through implementation, testing and debugging. The work also included preparation of Play-Store-ready builds, giving practical exposure beyond the implementation of individual gameplay features.

4.3 Problem Statement

Fruit Merge – Melon Maker: A fruit-merging puzzle requires smooth coordination between player input, fruit movement, physics-based collisions, fruit progression, scoring and game-state transitions. The challenge was to implement these systems so that the player could drop fruits into a confined play area and receive consistent feedback when matching fruits collided and merged.

Ball Sort: A ball-sorting puzzle requires controlled movement of colored balls between cups while maintaining valid game states, cup capacity, move counting, timing and level-completion conditions. The challenge was to provide a clear

interaction flow from selecting and moving a ball to validating a completed arrangement.

4.4 Project Objectives

1. Automatically provide a playable mobile puzzle experience for Fruit Merge – Melon Maker.
2. Implement fruit dropping, physics interaction and collision-driven merging.
3. Maintain score progression and player feedback in Fruit Merge.
4. Provide a complete game flow including main, gameplay and end states.
5. Implement Ball Sort as a mobile sorting puzzle with controlled ball movement.
6. Manage ball-selection, dragging, dropping and completion states.
7. Track moves, elapsed time and level progression in Ball Sort.
8. Provide pause, reset, game-over, score-submission and leaderboard-related screens.
9. Test and debug both games for reliable player interaction and game-state transitions.
10. Prepare mobile/Android-ready builds as part of the internship responsibilities.

4.5 Scope of the Project

The scope of Fruit Merge includes the Main, Game and End layouts; fruit objects from Fruit0 through Fruit9; physics-based movement and collision handling; score display; aiming and next-fruit interfaces; swap and blast add-ons; audio; and game-over/replay flow. The project is configured as a portrait-oriented mobile game with a 720×1280 resolution.

The scope of Ball Sort includes the Title Screen, Game, GameOver, SubmitScore, ViewLeaderboards, ScreenSaver and PauseMenu layouts. The project includes six ball variants, cup objects, gameplay states, move and time tracking, level progression, pause/reset controls, score-submission and leaderboard screens, audio, animations and visual effects.

4.6 Technologies and Tools Used

Technology / Tool	Purpose / Use
GDevelop 5.6	Game engine and event-based development environment used for both projects.
Physics2	Physics behaviour used for Fruit Merge fruit movement and collisions.
Bounce Extension	Extension included in the Fruit Merge project.
CrazyGames Ad API	Ad-related functionality included in the Fruit Merge project.

ExtendedMath	Mathematical/text-related functionality used in Ball Sort.
RepeatEveryXSeconds	Repeated timed actions used in Ball Sort.
Sticker	Visual/UI effect support used in Ball Sort.
CameraShake	Gameplay feedback effect used in Ball Sort.
CopyCameraSettings	Camera configuration support used in Ball Sort.
External Events	Reusable Ball Sort logic including GameLogic, DropBall, ExplodeCup and ButtonEffects.
Audio & Image Assets	Background music, sound effects, fruit, ball, cup, UI and visual-effect assets.

4.7 System Architecture

The two projects use event-driven game architecture. Game objects, layouts and event groups work together to process player input, update game variables, perform gameplay actions and transition between game states.

Fruit Merge – Architecture

Component	Role
Main Layout	Initial game presentation and transition into the gameplay layout.
Game Layout	Primary play area containing fruit objects, boundaries, score, aiming interface, next-fruit preview and add-ons.
Fruit Objects	Fruit0–Fruit9 represent the progression of fruit sizes/stages.
Physics2	Controls physical movement and collision behaviour of fruits.
Fruit Collision Events	Detect matching fruits, create the next fruit stage and update score.
Score & Coins	Maintains and displays the current score.
Out & Ads	Handles game-over/out conditions and ad-related functionality.
Sfx & Sounds	Controls background music and gameplay sound feedback.
End Layout	Displays the end state and replay flow.

Figure 4.1 shows the project identity asset used in the Fruit Merge project.

Figure 4.1: Fruit Merge – Melon Maker project identity.

Ball Sort – Architecture

Component	Role
Title Screen	Entry screen with game presentation and navigation.
Game Layout	Main sorting puzzle environment.
GameLogic	Reusable event logic for gameplay state flow and completion.
DropBall	Reusable ball-drop/movement logic.
ExplodeCup	Cup effect logic.
ButtonEffects	Reusable interface effects.
CurrentLevel / MovesMade / PlayTimer	Track level, moves and elapsed time.
MaxBallsPerCup	Controls the four-ball capacity of each cup.
GameOver / SubmitScore / ViewLeaderboards	Post-game, score submission and leaderboard-related flows.
PauseMenu	Pause, resume, reset-progress and navigation controls.

Figure 4.2 shows the supplied Ball Sort project thumbnail.

Figure 4.2: Ball Sort project thumbnail.

4.8 Methodology

The development methodology followed an iterative implementation-and-testing approach. Gameplay requirements were translated into layouts, objects, variables and event logic. After implementation, the games were tested through repeated player

interactions to identify incorrect state transitions, unexpected object behaviour and interface issues.

For Fruit Merge, the methodology focused on configuring the play area, fruit objects and physics, implementing matching-fruit collision events, updating the score, adding gameplay aids and validating the game-over/replay flow. The event structure separates fruit handling, collisions, score and coins, add-ons, sound and UI logic.

For Ball Sort, the methodology focused on a defined gameplay state flow: Idle, BallFloating, DraggingBall, DroppingBall, CheckIfComplete and LevelComplete. Reusable external events were used for game logic, ball dropping, cup effects and button effects. Testing included ball selection, dragging, valid/invalid drops, cup capacity, moves, timer, level completion and navigation screens.

4.9 Expected Outcomes

- A functional Fruit Merge mobile puzzle with fruit progression, scoring, physics and game-over flow.
- A functional Ball Sort mobile puzzle with ball sorting, state management, level progression, moves and timing.
- Improved practical understanding of event-driven game development using GDevelop.
- Hands-on experience with game testing, debugging, UI, audio, animation and visual effects.
- Practical exposure to Android and Play-Store-ready build preparation.
- Development experience that connects academic programming concepts with interactive software applications.

4.10 Certificates of Completion and Other Communication Mail Proof

The internship completion certificate issued by IBRA Digital Branding Services Pvt. Ltd. is attached in this report as proof of the internship. The certificate confirms the internship period of 15 April 2026 to 15 August 2026, the designation of Game Developer Intern, the Game Development Department, and the work involving Android mobile games, functionality, testing, debugging and Play-Store-ready builds. Any additional communication or email proof required by the university can be inserted in this section before final submission.

5. BIBLIOGRAPHY / REFERENCES

1. GDevelop Documentation. GDevelop game-development documentation and project resources.
2. GDevelop project file: Fruit Merge – Melon Maker Game. Submitted project files used as the primary source for project structure and implementation details.
3. GDevelop project file: IBRA_BallSort. Submitted project files used as the primary source for project structure and implementation details.
4. Internship Completion Certificate issued by IBRA Digital Branding Services Pvt. Ltd., confirming internship period, designation, department and broad responsibilities.
5. IILM University, Internship Report Format updated 26–27. University-provided formatting and structural guidelines.